

Chapter 3

Image Degradation, Restoration and Reconstruction

IMP302m

Contents

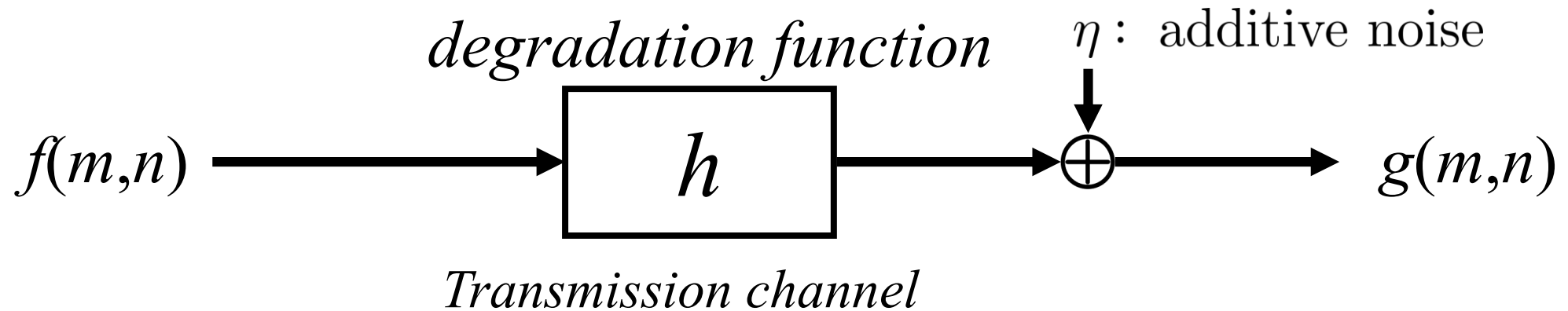
- 3.1 A model of image restoration process
- 3.2 Noise models
- 3.3 Spatial filtering and frequency domain filtering in noise reduction
- 3.4 Degradation and its function
- 3.5 Wavelet Denoising for Image Enhancement
- 3.6 Minimum mean square error filtering
- 3.7 Constrained least square filtering
- 3.8 Basic Methods for Image Restoration and Identification
- 3.9 Image reconstruction from projections
- 3.10 Multichannel Image Recovery

3.1 A model of image restoration process

- The principal goal of restoration techniques is to improve an image in some predefined sense
- Restoration attempts to recover an image that has been degraded by using a priori knowledge of the degradation phenomenon
- Oriented toward modeling the degradation and applying the inverse process in order to recover the original image

3.1 A model of image restoration process

- Image transmitted over channel is degraded

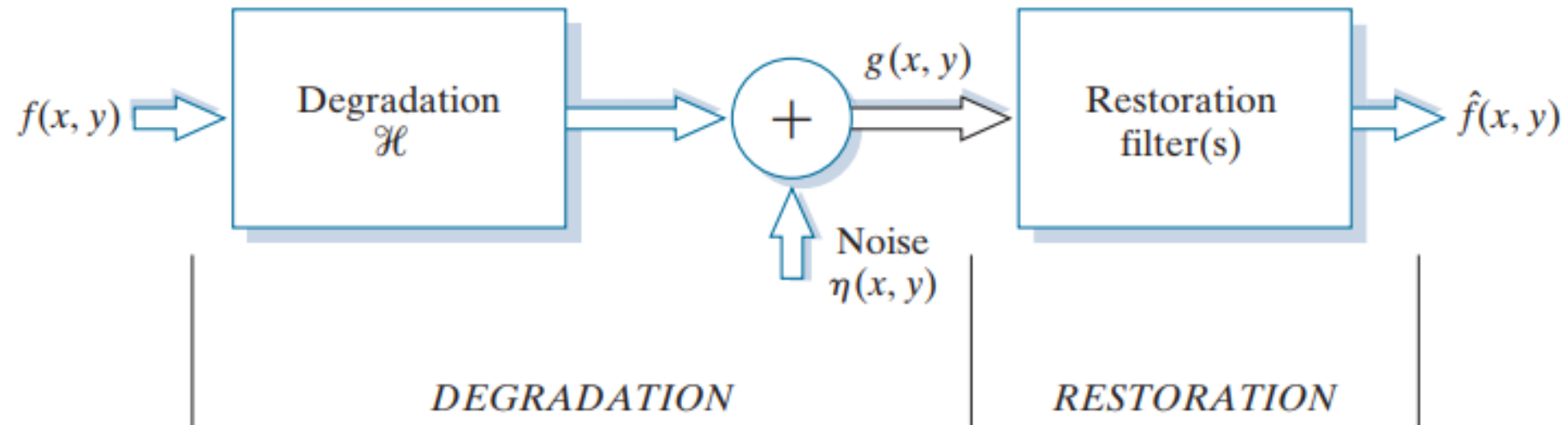


$$g(x, y) = (h \star f)(x, y) + \eta(x, y)$$

- In frequency domain: $G(u, v) = H(u, v)F(u, v) + N(u, v)$
- Restoration: estimate to be as close as possible to the original image

3.1 A model of image restoration process

- Full process to restore a transmitted image



3.2 Noise models

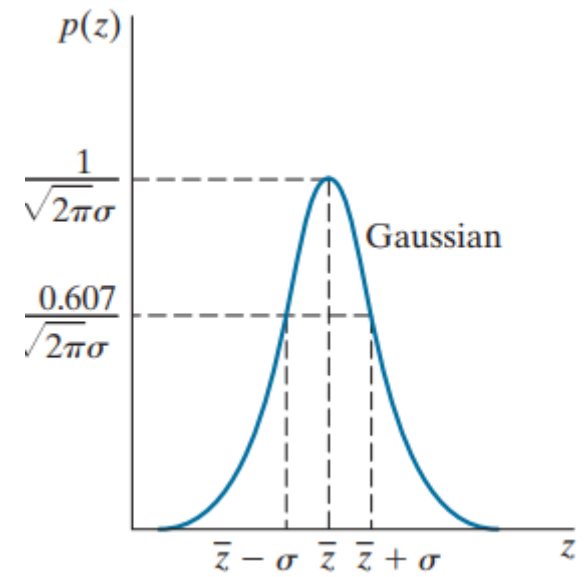
- The principal sources of noise in digital images arise during image acquisition and/or transmission
 - Example: an image transmitted using a wireless network might be corrupted by lightning or other atmospheric disturbance.
- When the Fourier spectrum of noise is constant, the noise is called *white noise*
- Noise is considered as independent of spatial coordinates, and that it is uncorrelated with respect to the image itself
 - Example: quantum-limited imaging, such as in X-ray and nuclear-medicine imaging

3.2 Noise models- Probability density functions (PDF)

- Gaussian Noise

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\bar{z})^2}{2\sigma^2}}$$

z represents intensity, $\bar{z}$ is the mean (average) value of z , and σ is its standard deviation



3.2 Noise models- Probability density functions (PDF)

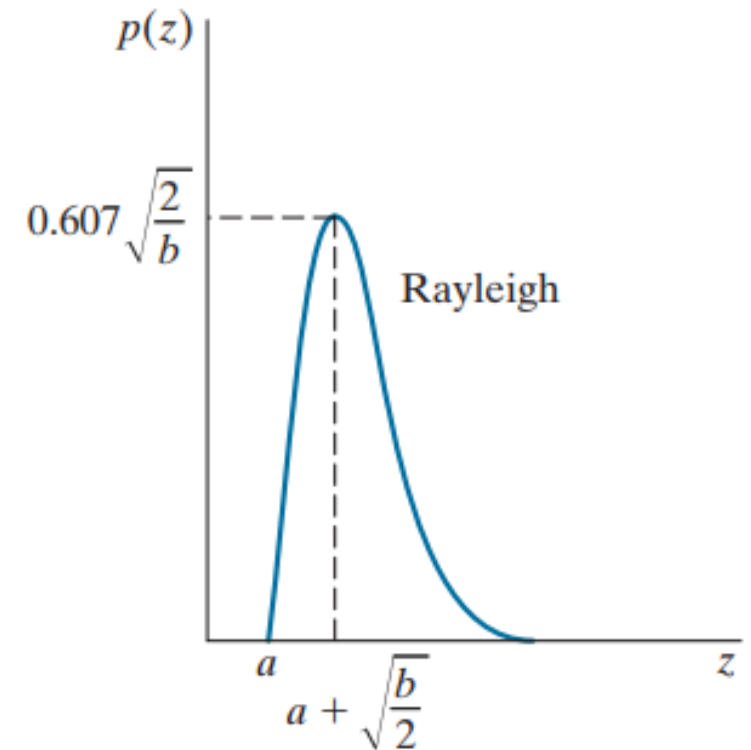
- Rayleigh Noise

$$p(z) = \begin{cases} \frac{2}{b}(z-a)e^{-\frac{(z-a)^2}{b}} & z \geq a \\ 0 & z < a \end{cases}$$

- The mean and variance of z when this random variable is characterized by a Rayleigh PDF are

$$\bar{z} = a + \sqrt{\pi b/4}$$

$$\sigma^2 = \frac{b(4 - \pi)}{4}$$



3.2 Noise models- Probability density functions (PDF)

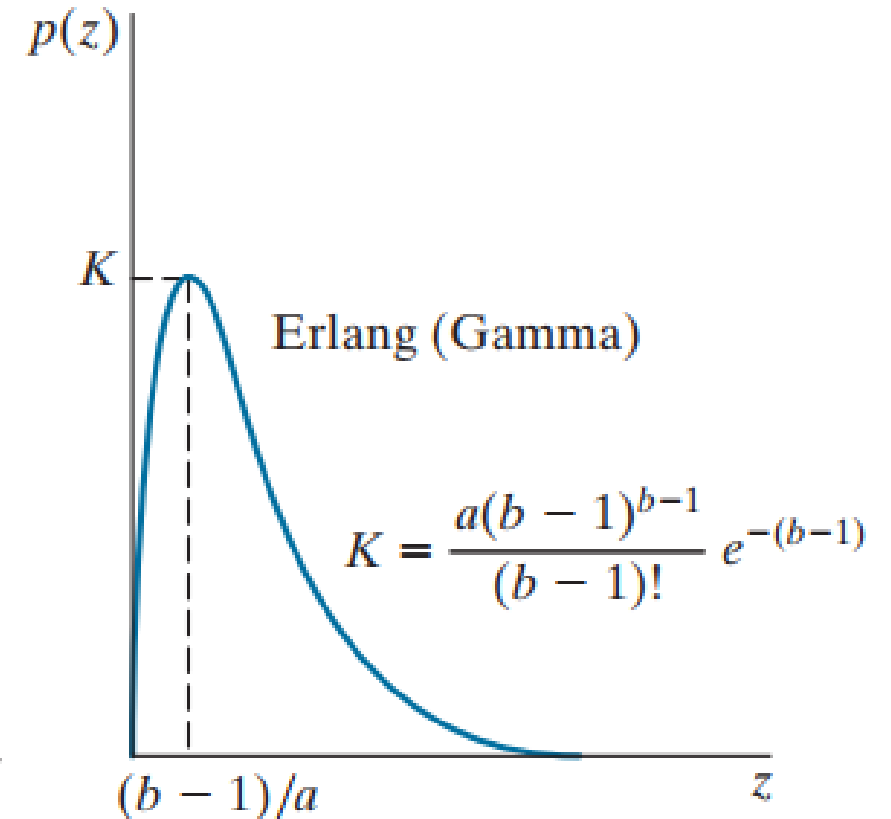
- Erlang (Gamma) Noise

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

- $a > b$ and b is a positive integer
- The mean and variance of z

$$\bar{z} = \frac{b}{a}$$

$$\sigma^2 = \frac{b}{a^2}$$



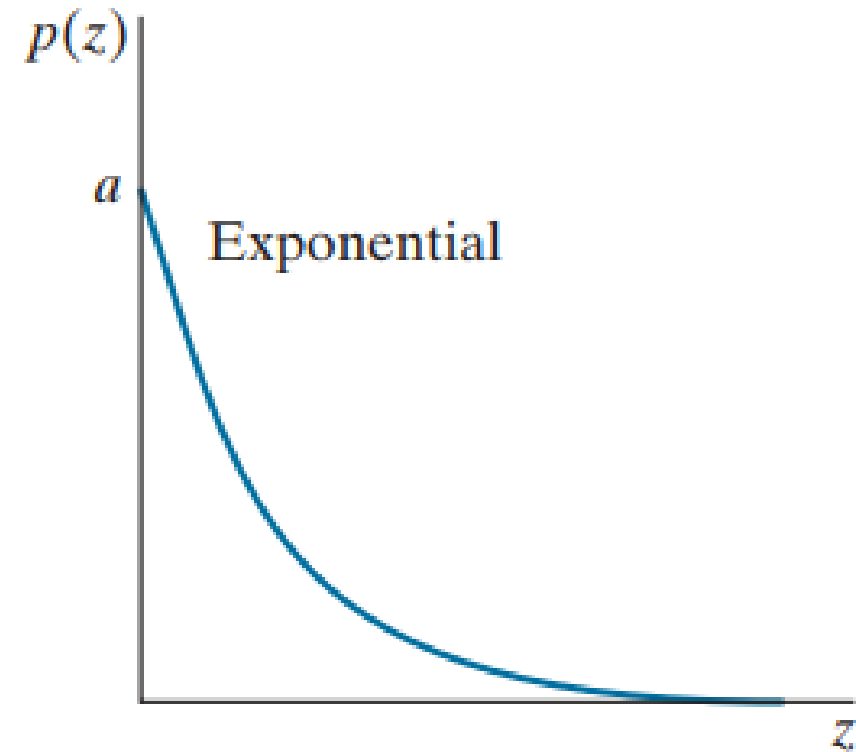
3.2 Noise models- Probability density functions (PDF)

- Exponential Noise

$$p(z) = \begin{cases} ae^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

- $a > 0$
- The mean and variance of z

$$\bar{z} = \frac{1}{a} \quad \sigma^2 = \frac{1}{a^2}$$



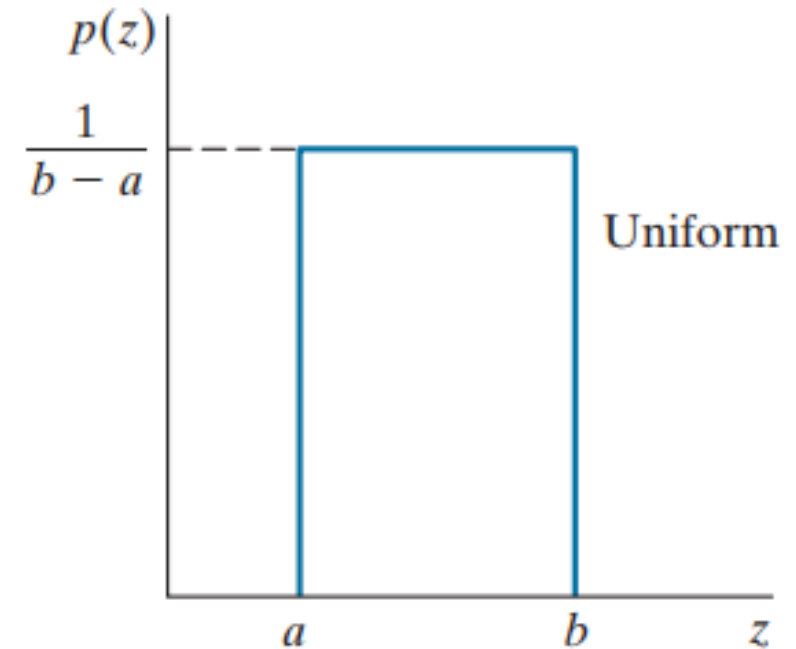
3.2 Noise models- Probability density functions (PDF)

- Uniform Noise

$$p(z) = \begin{cases} \frac{1}{b-a} & a \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$

- The mean and variance of z

$$\bar{z} = \frac{a+b}{2} \quad \sigma^2 = \frac{(b-a)^2}{12}$$

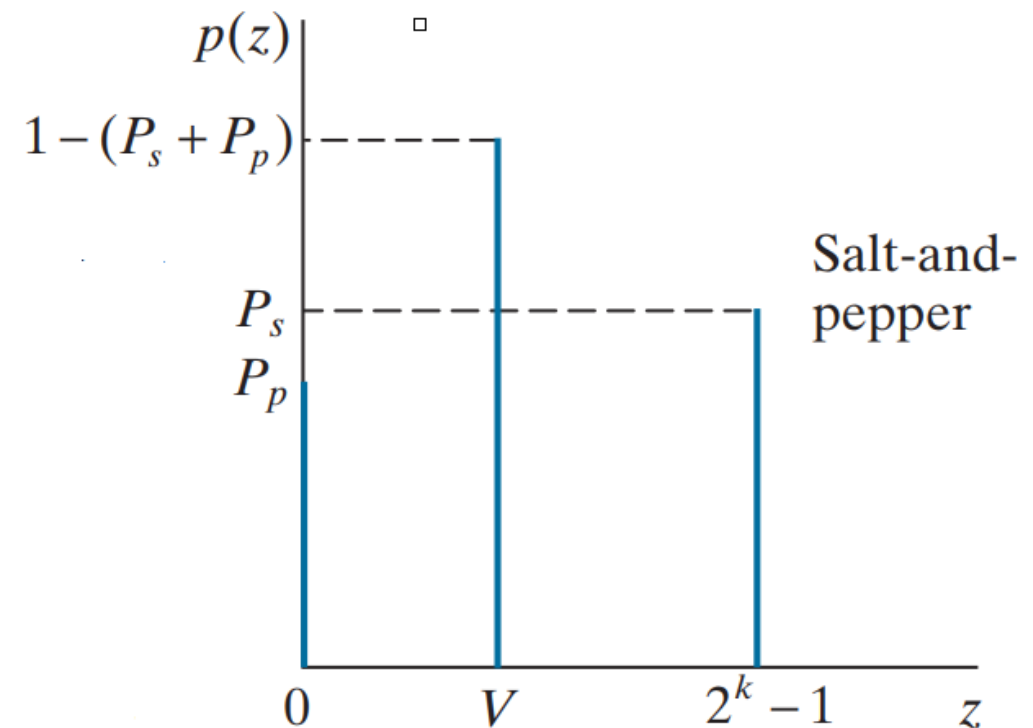


3.2 Noise models- Probability density functions (PDF)

- Salt-and-Pepper Noise

$$p(z) = \begin{cases} P_s & \text{for } z = 2^k - 1 \\ P_p & \text{for } z = 0 \\ 1 - (P_s + P_p) & \text{for } z = V \end{cases}$$

- V is any integer value in the range $0 < V < 2^k - 1$



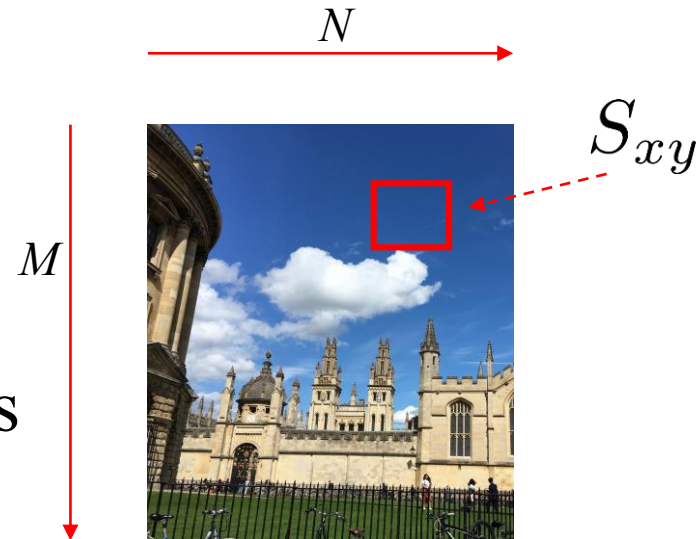
3.3 Spatial filtering and frequency domain filtering in noise reduction

Spatial filtering – Mean filter

- rectangular subimage window (neighborhood) of size $m \times n$
- the corrupted image, $g(x, y)$, in the area defined by S_{xy}
- Arithmetic Mean Filter

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(r, c) \in S_{xy}} g(r, c)$$

r and c are the row and column coordinates of the pixels contained in the neighborhood S_{xy}



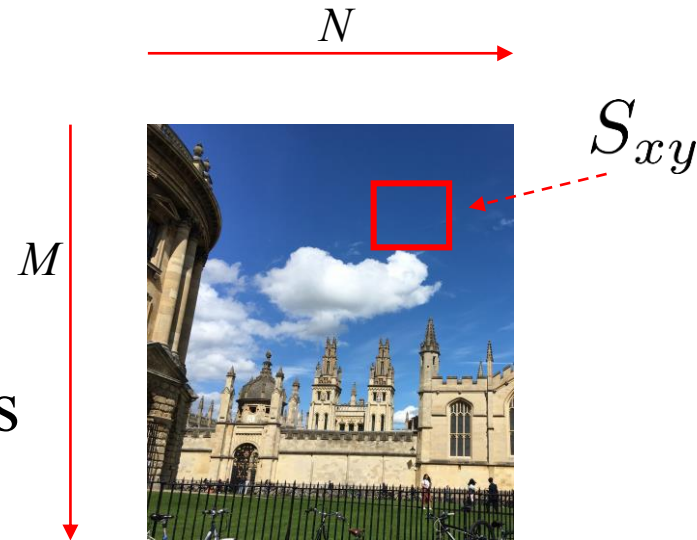
3.3 Spatial filtering and frequency domain filtering in noise reduction

Spatial filtering – Mean filter

- rectangular subimage window (neighborhood) of size $m \times n$
- the corrupted image, $g(x, y)$, in the area defined by S_{xy}
- Geometric Mean Filter

$$\hat{f}(x, y) = \left[\prod_{(r, c) \in S_{xy}} g(r, c) \right]^{\frac{1}{mn}}$$

r and c are the row and column coordinates of the pixels contained in the neighborhood S_{xy}



Geometric mean filter achieves smoothing comparable to arithmetic mean filter, but it tends to lose less image detail in the process!

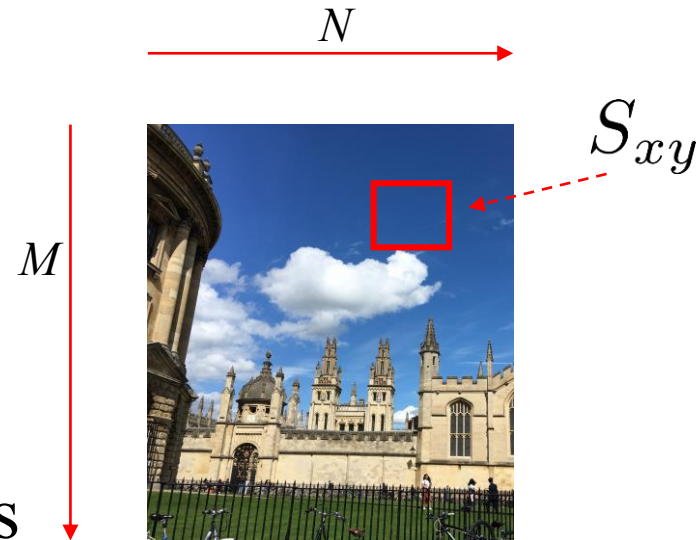
3.3 Spatial filtering and frequency domain filtering in noise reduction

Spatial filtering – Mean filter

- rectangular subimage window (neighborhood) of size $m \times n$
- the corrupted image, $g(x, y)$, in the area defined by S_{xy}
- Harmonic Mean Filter

$$\hat{f}(x, y) = \frac{mn}{\sum_{(r,c) \in S_{xy}} \frac{1}{g(r, c)}}$$

r and c are the row and column coordinates of the pixels contained in the neighborhood S_{xy}



harmonic mean filter works well for salt noise, Gaussian noise,
but fails for pepper noise!

3.3 Spatial filtering and frequency domain filtering in noise reduction

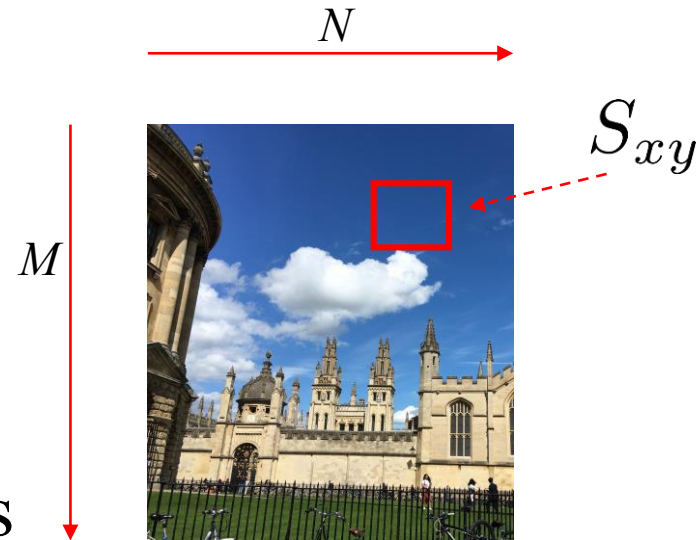
Spatial filtering – Mean filter

- rectangular subimage window (neighborhood) of size $m \times n$
- the corrupted image, $g(x, y)$, in the area defined by S_{xy}
- Contraharmonic Mean Filter

$$\hat{f}(x, y) = \frac{\sum_{(r, c) \in S_{xy}} g(r, c)^{Q+1}}{\sum_{(r, c) \in S_{xy}} g(r, c)^Q}$$

r and c are the row and column coordinates of the pixels contained in the neighborhood S_{xy} , Q is the order

harmonic mean filter works well for salt noise, Gaussian noise,
but fails for pepper noise!



3.3 Spatial filtering and frequency domain filtering in noise reduction

- Discussion: How do you think of adaptive noise reduction filter?

3.3 Spatial filtering and frequency domain filtering in noise reduction

Frequency domain filtering

- Periodic noise can be analyzed and filtered quite effectively using frequency domain techniques.
- The basic idea is that periodic noise appears as concentrated bursts of energy in the Fourier transform, at locations corresponding to the frequencies of the periodic interference.
- In restoration of images corrupted by periodic interference, the tool of choice is a notch filter

3.3 Spatial filtering and frequency domain filtering in noise reduction

- Frequency domain filtering – Notch filtering
- **Notch reject** filter transfer functions are constructed as products of highpass filter transfer functions whose centers have been translated to the centers of the notches.

$$H_{\text{NR}}(u, v) = \prod_{k=1}^Q H_k(u, v) H_{-k}(u, v)$$

where $H_k(u, v)$ and $H_{-k}(u, v)$ are highpass filter transfer functions whose centers are at (u_k, v_k) and $(-u_k, -v_k)$. Q is the number of notch filter pairs

3.3 Spatial filtering and frequency domain filtering in noise reduction

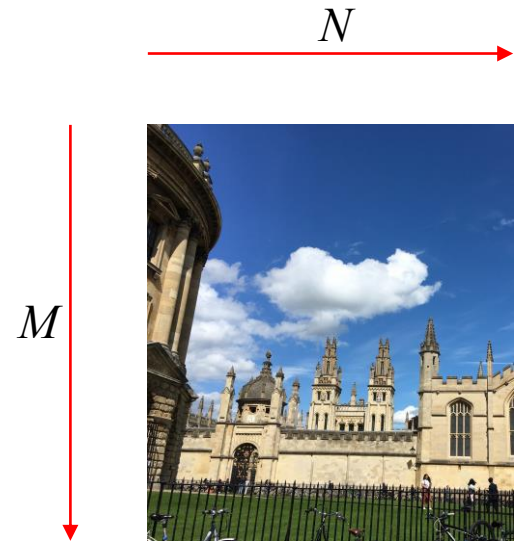
- These centers are specified with respect to the center of the frequency rectangle, $[\text{floor}(M/2), \text{floor}(N/2)]$
- The distance computations for the filter transfer functions

$$D_k(u, v) = \left[(u - M/2 - u_k)^2 + (v - N/2 - v_k)^2 \right]^{1/2}$$

$$D_{-k}(u, v) = \left[(u - M/2 + u_k)^2 + (v - N/2 + v_k)^2 \right]^{1/2}$$

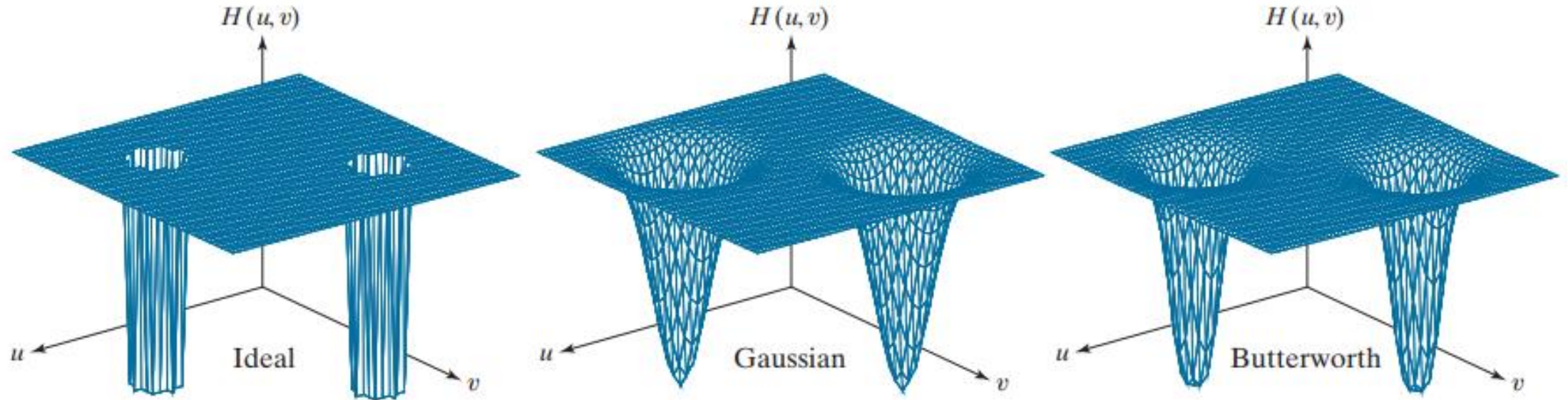
- **Notch pass** filter transfer function is obtained from a notch reject function using the expression

$$H_{\text{NP}}(u, v) = 1 - H_{\text{NR}}(u, v)$$



3.3 Spatial filtering and frequency domain filtering in noise reduction

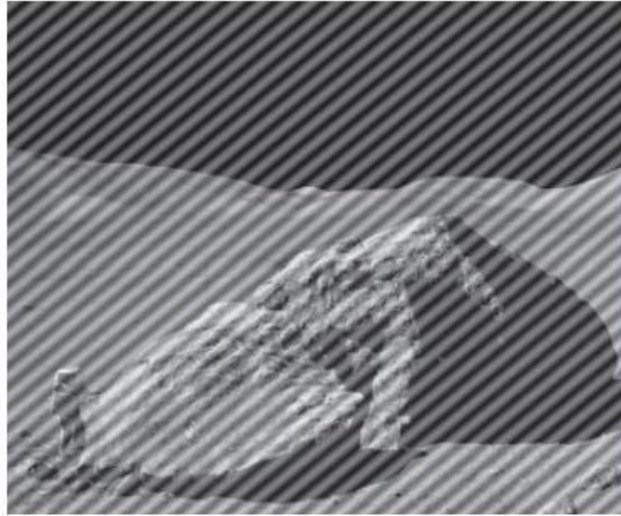
- Three typical notch reject filter transfer functions



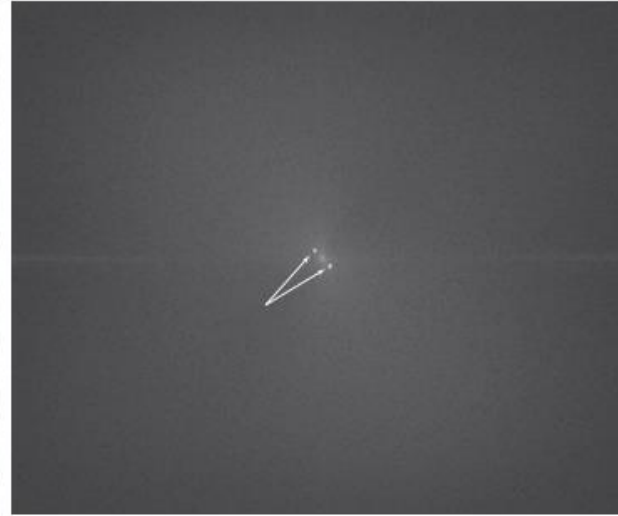
3.3 Spatial filtering and frequency domain filtering in noise reduction

- Example

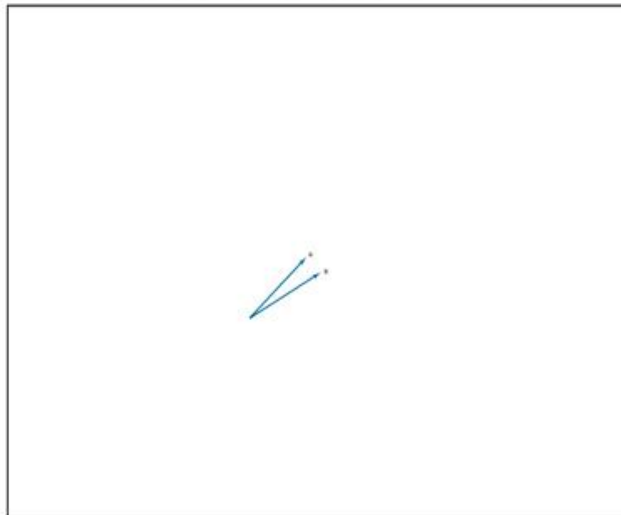
(a) Image corrupted by sinusoidal interference



(b) spectrum



(c) Notch reject filter

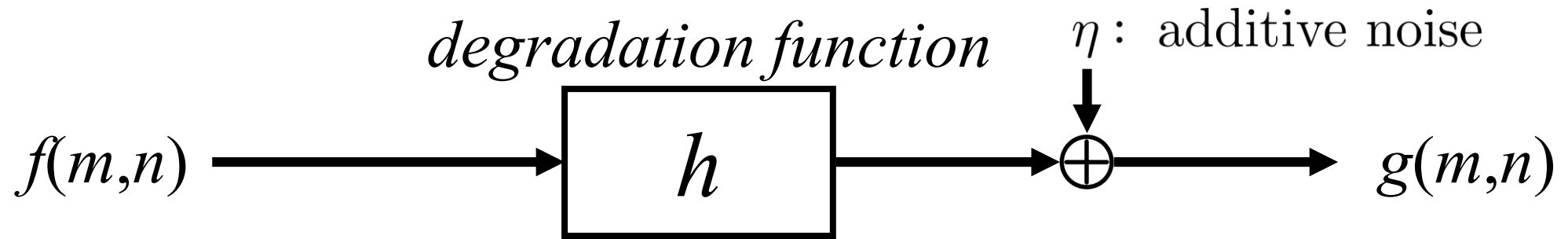


(d) Result



3.4 Degradation and its function

- Linear, position-invariant degradations



$$g(x, y) = \mathcal{H} [f(x, y)] + \eta(x, y)$$

- Degradation function is linear if

$$\mathcal{H} [a f_1(x, y) + b f_2(x, y)] = a \mathcal{H} [f_1(x, y)] + b \mathcal{H} [f_2(x, y)]$$

3.4 Degradation and its function

- *additivity*
- *homogeneity*
- *position (or space) invariant*

3.4 Degradation and its function

- *Impulse response* of degradation function

$$h(x, \alpha, y, \beta) = \mathcal{H} [\delta(x - \alpha, y - \beta)]$$

- *Superposition (or Fredholm) integral of the first kind*

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) h(x, \alpha, y, \beta) d\alpha d\beta$$

- Degradation function is position invariant in the presence of noise

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) h(x - \alpha, y - \beta) d\alpha d\beta + \eta(x, y)$$

- In frequency domain

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

3.4 Degradation and its function – Estimating the degradation function

- Estimating the degradation function is crucial for effective image restoration.
- Accurate estimation leads to better recovery of the original image
- Principal estimation methods:
 - Observation: Directly observing the degradation process
 - Experimentation: Conducting experiments to determine the degradation characteristics.
 - Mathematical modeling: Using mathematical models to estimate the degradation function based on assumptions and prior knowledge.
- The process of restoring an image by using a degradation function that has been estimated by any of these approaches sometimes is called *blind deconvolution*

3.4 Degradation and its function – Estimating the degradation function

Estimation by image observation

- Let the observed subimage be denoted by $g_s(x, y)$ and let the processed subimage (which in reality is our estimate of the original image in that area) be denoted by $\hat{f}_s(x, y)$

$$\hat{H}_s(u, v) = \frac{G_s(u, v)}{\hat{F}_s(u, v)}$$

- Then $\hat{H}(u, v)$ can be derived based on the property of space invariant

3.4 Degradation and its function – Estimating the degradation function

Estimation by experimentation

- The degradation function can be estimated as

$$\hat{H}_s(u, v) = \frac{G_s(u, v)}{A}$$

- A is a constant describing the strength of the impulse

3.4 Degradation and its function – Estimating the degradation function

Estimation by modeling

- The degradation function can be estimated using the following model

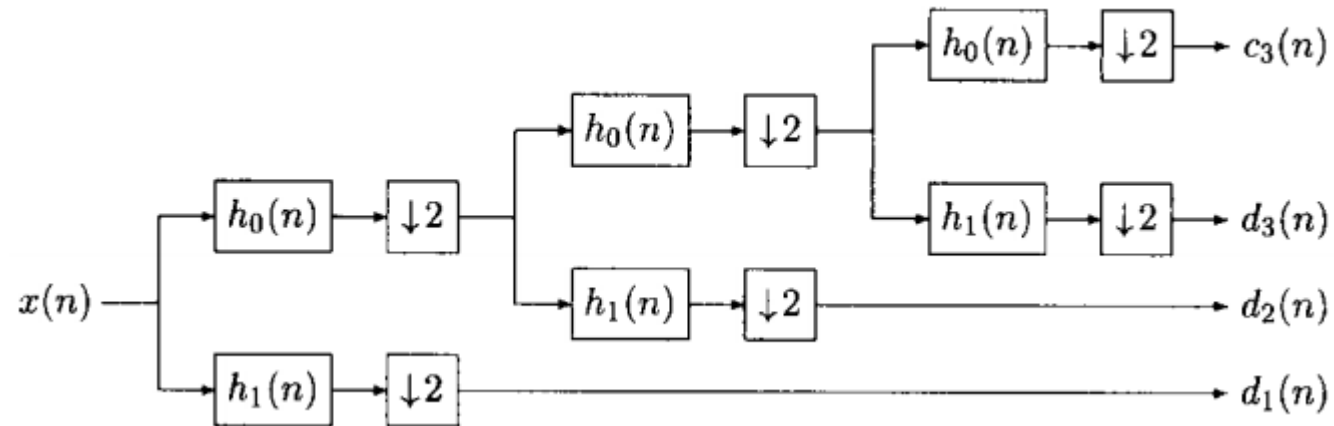
$$H_s(u, v) = e^{-k(u^2 + v^2)^{5/6}}$$

where k is a constant that depends on the nature of the turbulence

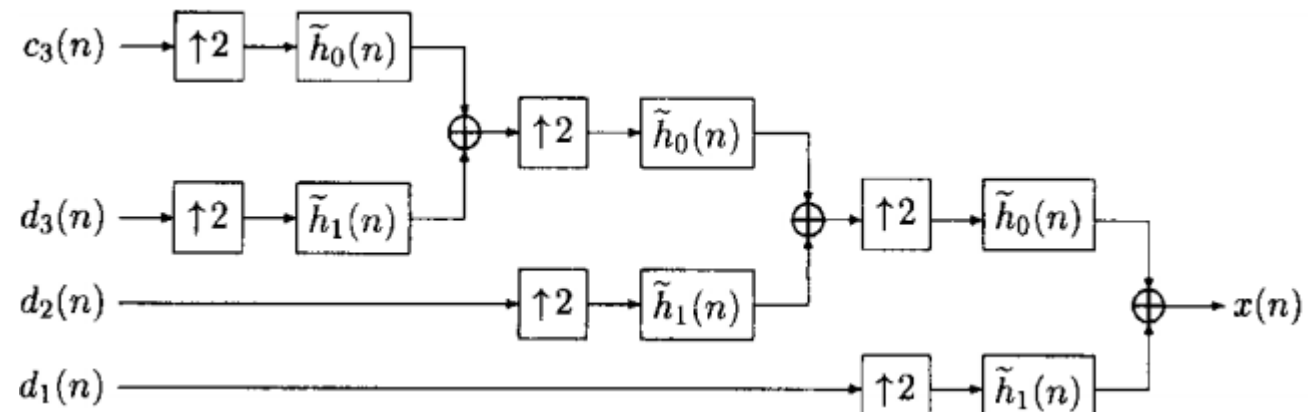
- It is nearly same as Gaussian lowpass filter

3.5 Wavelet Denoising for Image Enhancement

- The discrete wavelet transform (DWT) is a multiresolution (or multiscale) representation. The DWT is implemented via multirate filterbanks.



A three-level forward DWT via a two-channel iterative filterbank



A three-level inverse DWT via a two-channel iterative filterbank

3.5 Wavelet Denoising for Image Enhancement – Main steps

- Decompose the image into multiple levels of wavelet coefficients using the Discrete Wavelet Transform (DWT)
- The image is represented at different scales, with high-frequency components (details) and low-frequency components (approximation)
- Apply thresholding to the wavelet coefficients to suppress noise
 - Soft thresholding: reduces all coefficients by a threshold value, setting those below the threshold to zero
 - Hard thresholding: sets coefficients below the threshold to zero while keeping others unchanged
 - Discussion: Adaptive thresholding?
- Reconstruct the denoised image by applying the inverse DWT to the thresholded coefficients

3.6 Minimum mean square error filtering

- The error measure between estimated image $\hat{f}$ and original image f is defined as

$$e^2 = \mathbb{E} \{ (f - \hat{f})^2 \}$$

- Objective?

3.6 Minimum mean square error filtering

- Objective: Find an estimate $\hat{f}$ of the uncorrupted image f such that the mean square error between them is minimized.

$$\hat{F} = \arg \min_{\hat{f}} e^2 = \arg \min_{\hat{f}} \mathbb{E} \{ (f - \hat{f})^2 \}$$

- The noise and the image are uncorrelated, each has zero mean, the intensity levels in the estimate are a linear function of the levels in the degraded image → Estimated image in frequency domain

$$\hat{F}(u, v) = \left[\frac{1}{H(u, v)} \frac{|H(u, v)|^2}{|H(u, v)|^2 + S_{\eta}(u, v)/S_f(u, v)} \right] G(u, v)$$

Wiener filter/minimum mean square error filter/least square error filter

3.6 Minimum mean square error filtering

- Measures to characterize the performance of restoration algorithms – signal-to-noise ratio

$$\text{SNR} = \frac{\sum_{u=0}^{M-1} \sum_{v=0}^{N-1} |F(u, v)|^2}{\sum_{u=0}^{M-1} \sum_{v=0}^{N-1} |N(u, v)|^2}$$

- Mean square error

$$\text{MSE} = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x, y) - \hat{f}(x, y)]^2$$

3.7 Constrained least square filtering

- Difficulty in Wiener filter?
 - the power spectra of the undegraded image
 - based on minimizing a statistical criterion and, as such, it is optimal in an average sense

Is there a way to yield an optimal result for each image to which it is applied?

3.7 Constrained least square filtering

- Spatial domain the degraded image is

$$g(x, y) = (h \star f)(x, y) + \eta(x, y)$$

- In matrix form

$$\mathbf{g} = \mathbf{H} \mathbf{f} + \boldsymbol{\eta}$$

- Reduce the effects of noise sensitivity: Laplacian $\rightarrow$ find the minimum of a criterion function, C

$$C = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\nabla^2 f(x, y)]^2$$

subject to

$$\|\mathbf{g} - \mathbf{H} \hat{\mathbf{f}}\|^2 = \|\boldsymbol{\eta}\|^2$$

3.7 Constrained least square filtering

- The frequency domain solution

$$\hat{F}(u, v) = \left[\frac{H^*(u, v)}{|H(u, v)|^2 + \gamma |P(u, v)|^2} \right] G(u, v)$$

where γ is a parameter that must be adjusted so that the constraint is satisfied, and $P(u, v)$ is the Fourier transform of the function

$$p(u, v) = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

3.7 Constrained least square filtering

- Loop for determining γ , what we want $\|\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\|^2 = \|\boldsymbol{\eta}\|^2 \pm \alpha$
 1. Specify an initial value of γ
 2. Compute $\|\mathbf{r}\|^2 = \|\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\|^2$
 3. Stop if condition (what we want) is satisfied; otherwise return to Step 2 after increasing γ if $\|\mathbf{r}\|^2 < \|\boldsymbol{\eta}\|^2 - \alpha$ or decreasing γ if $\|\mathbf{r}\|^2 > \|\boldsymbol{\eta}\|^2 + \alpha$. Use the new value of γ to recompute the optimum estimate $\hat{F}(u, v)$

3.8 Basic Methods for Image Restoration and Identification

- Inverse Filter (with ideal filter)

$$\hat{F}(u, v) = \frac{1}{D(u, v)} (D(u, v)F(u, v) + N(u, v)) = F(u, v) + \frac{N(u, v)}{D(u, v)}$$

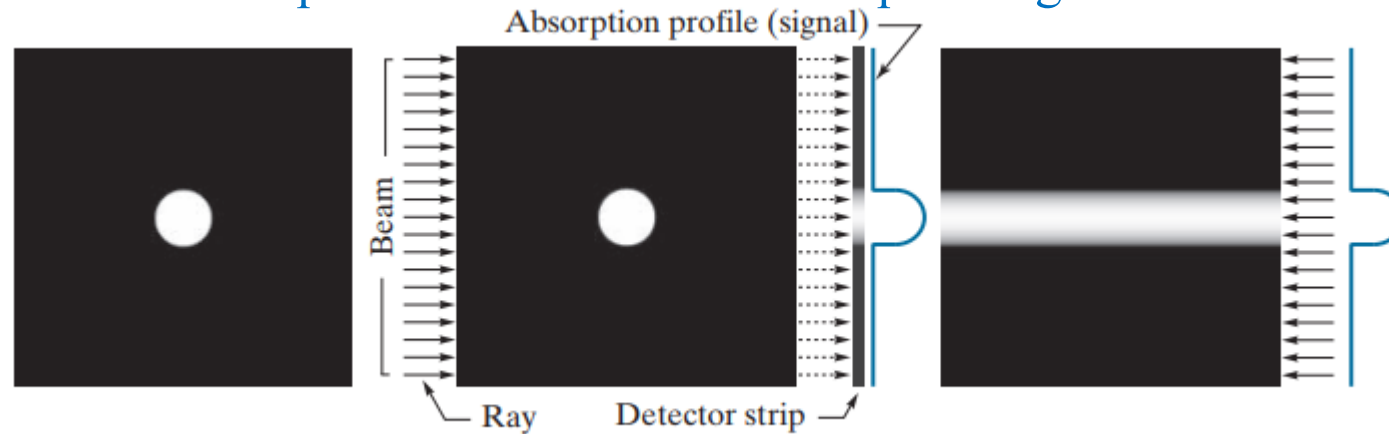
- Wiener filter (with ideal filter)

$$\hat{F}(u, v) = \left[\frac{1}{D(u, v)} \frac{|D(u, v)|^2}{|D(u, v)|^2 + S_\eta(u, v)/S_f(u, v)} \right] G(u, v)$$

3.9 Image reconstruction from projections

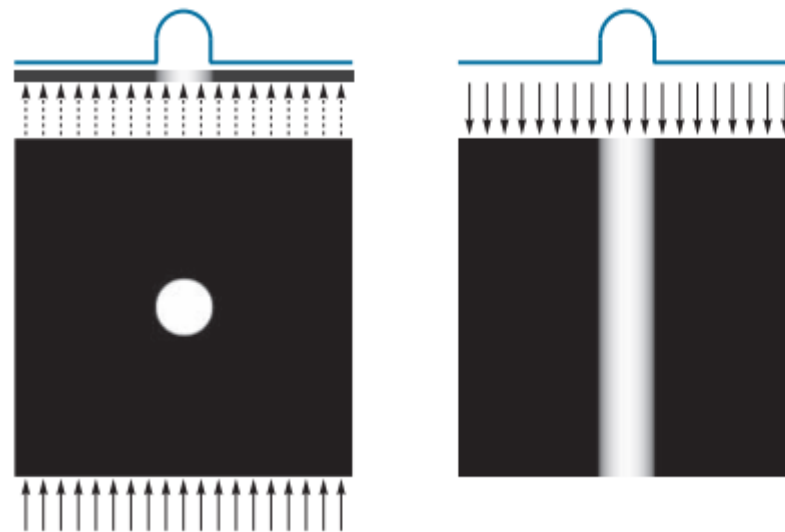
(b) Parallel beam, detector strip and profile of sensed 1-D absorption signal

(a) Flat region with single object



(c) Result of back projecting the absorption profile

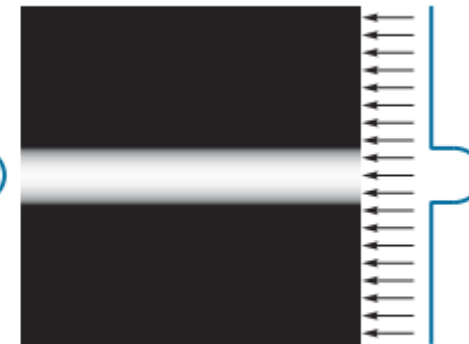
(d) Meab and detectors rotated by 90 degrees



(e) backprojection



(f) Sum of c and e



3.9 Image reconstruction from projections

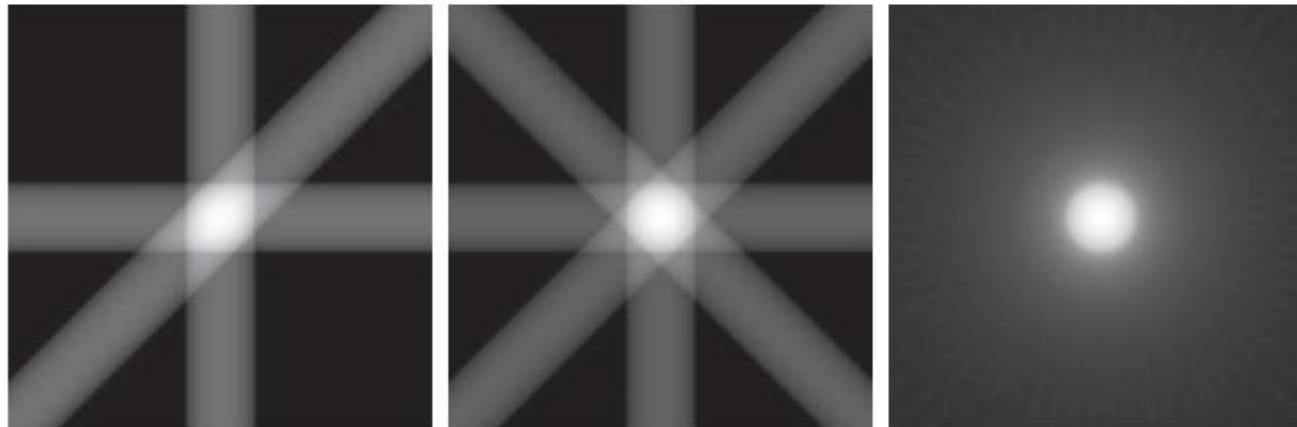
(b) Reconstruction
using 1 backprojection

(a) Flat region
with single object



(c) Reconstruction using
2 backprojections

(d) Reconstruction
using 3 backprojections



(f) Reconstruction using
32 backprojections

(e) Reconstruction using
4 backprojections

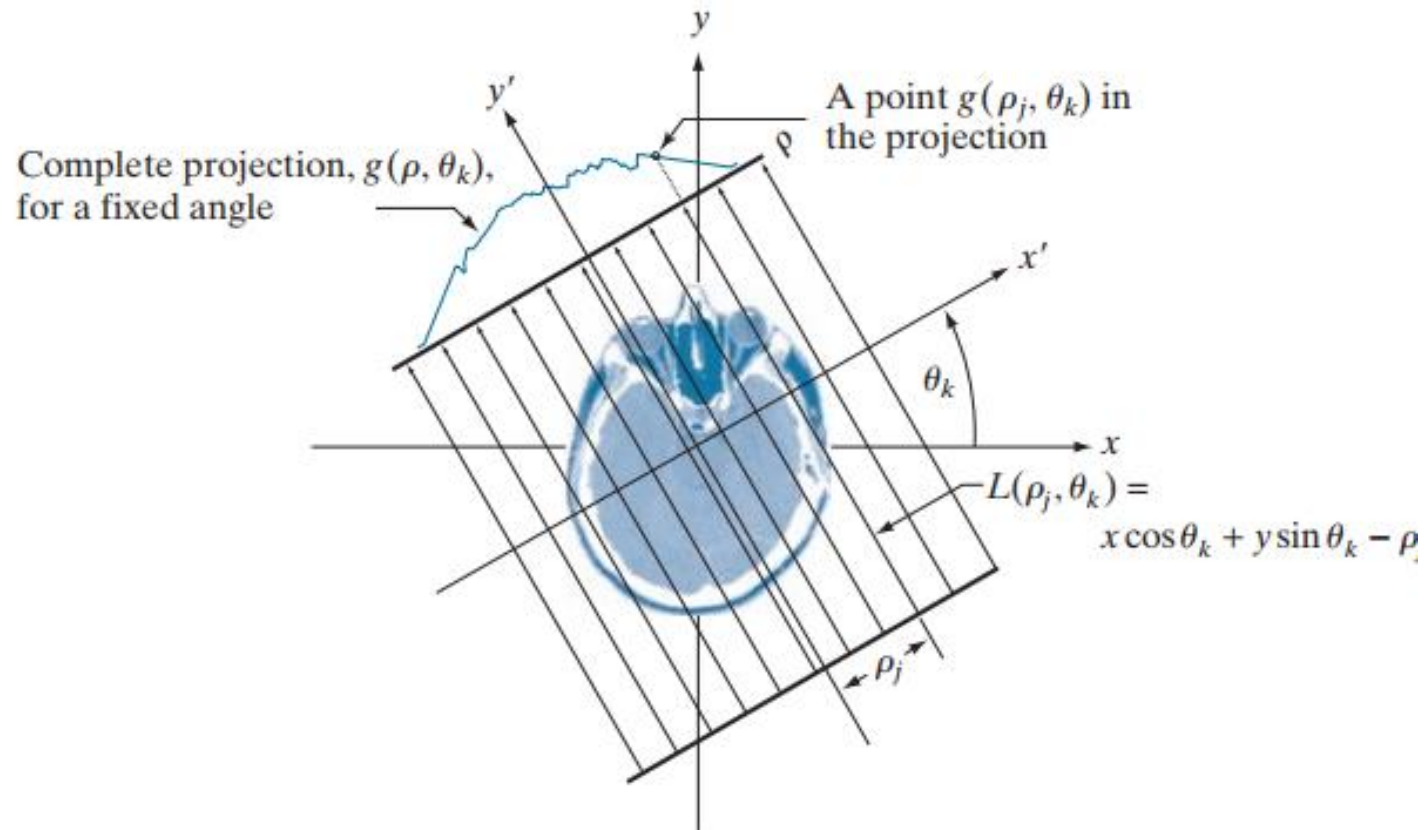
3.9 Image reconstruction from projections -

Application: Principles of X-ray computed tomography (CT)

- The goal of X-ray computed tomography is to obtain a 3-D representation of the internal structure of an object by X-raying the object from many different directions
- Computed tomography attempts to get that same information (or localized parts of it) by generating slices through the body. A 3-D representation then can be obtained by stacking the slices

3.9 Image reconstruction from projections -

Application: Principles of X-ray computed tomography (CT)



3.10 Multichannel Image Recovery

- Multichannel images
 - Color images, video frames, medical imaging, multispectral satellite images
- Caused
 - Multiple image frames or channels
 - Depict the same scene or object acquired by different sensors or at different times

3.10 Multichannel Image Recovery

- Image recovery
 - Ambiguities in inferring the desired image from the observations usually arise from uncertainty produced by noise → reduced if, in addition to information provided by the observed data, one also has prior knowledge about the desired image.
- Advantages in processing multichannel images
 - Multichannel images offer the possibility of exploiting correlations *between* the channels in addition to those within each channel.
 - By utilizing this extra information, multichannel image recovery can yield tremendous benefits over separate recovery of the component channels.

3.10 Multichannel Image Recovery – Image model

- Degraded multichannel image (K-channel)

$$\mathbf{g} = \mathbf{H}\mathbf{f} + \boldsymbol{\eta}$$

$$\begin{bmatrix} g_1 \\ g_2 \\ \vdots \\ g_K \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} & \cdots & H_{1K} \\ H_{21} & H_{22} & \cdots & H_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ H_{K1} & H_{K2} & \cdots & H_{KK} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_K \end{bmatrix} + \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_K \end{bmatrix}$$

3.10 Multichannel Image Recovery – Multichannel image estimation approaches

- Linear Minimum Mean Square Error (LMMSE) Estimation

$$\mathbf{f}_{\text{LMMSE}} = \mathbf{C}_f \mathbf{H}^T (\mathbf{H} \mathbf{C}_f \mathbf{H}^T + \mathbf{C}_\eta)^{-1} \mathbf{g}$$

where the covariance matrix of the multichannel noise vector

$$\mathbf{C}_\eta = \mathbb{E} \{ \boldsymbol{\eta} \boldsymbol{\eta}^T \}$$

and the covariance matrix of the multichannel image vector

$$\mathbf{C}_f = \mathbb{E} \{ \mathbf{f} \mathbf{f}^T \}$$

3.10 Multichannel Image Recovery – Multichannel image estimation approaches

Regularized Weighted Least-Squares (RWLS) Estimation

- Weighted Least-Squares (WLS) estimate of $\mathbf{f}$

$$\mathbf{f}_{\text{WLS}} = \arg \min_{\mathbf{f}} \{ (\mathbf{H}\mathbf{f} - \mathbf{g})^T \mathbf{C}_{\eta}^{-1} (\mathbf{H}\mathbf{f} - \mathbf{g}) \}$$

- Regularized Weighted Least-Squares (RWLS) Estimation

$$\mathbf{f}_{\text{RWLS}} = \arg \min_{\mathbf{f}} \{ (\mathbf{H}\mathbf{f} - \mathbf{g})^T \mathbf{C}_{\eta}^{-1} (\mathbf{H}\mathbf{f} - \mathbf{g}) + \lambda \|\mathbf{Q}\mathbf{f}\|^2 \}$$

where λ the regularization parameter, controls the trade-off between fidelity to the data and smoothness of the estimate, $\mathbf{Q}$ is highpass filter

- Solution

$$\mathbf{f}_{\text{RWLS}} = (\mathbf{H}^T \mathbf{C}_{\eta}^{-1} \mathbf{H} + \lambda \mathbf{Q}^T \mathbf{Q})^{-1} \mathbf{H}^T \mathbf{C}_{\eta}^{-1} \mathbf{g}$$

3.11 Motion Detection and Estimation - Motion Detection

- Motion detection: Identifying moving regions or points between two time instants in images.
- Applicable primarily to static cameras.
- Requires counteracting camera motion through global motion estimation and compensation for moving camera
- Challenges
 - Motion detection relies on intensity changes, not directly on the motion of image points.
 - Intensity changes can be caused by camera noise, illumination variations, object motion (which might not change intensity noticeably).

3.11 Motion Detection and Estimation - Motion Detection

- In moving areas the difference of the pixel intensity at two frames/times/.. is attributed to motion

$$\rho_k[\mathbf{n}] = I_k[\mathbf{n}] - I_{k-1}[\mathbf{n}]$$

- Hypothesis testing with fixed threshold

$$\rho_k^2[\mathbf{n}] \underset{\mathcal{S}}{\overset{\mathcal{M}}{\gtrless}} \theta$$

where $\mathcal{M}$ is moving and $\mathcal{S}$ is stationary

- To attenuate the impact of noise, averaging the temporal differences over an N -point spatial window W_N , centered at $\mathbf{n}$, can be used:

$$\frac{1}{N} \sum_{m \in W_N} \rho_k^2[\mathbf{m}] \underset{\mathcal{S}}{\overset{\mathcal{M}}{\gtrless}} \theta \text{ -----> Adaptive threshold?}$$

3.11 Motion Detection and Estimation - Motion Estimation

- Understanding motion is important for both video compression and processing
- Video compression techniques explicitly reduce the number of bits needed to represent a video sequence, with motion-compensated hybrid coding
 - Used in video compression standards (H.261, H.263, H.264, MPEG-1, MPEG-2, MPEG-4)
- Video processing methods transform the video sequence to improve quality without reducing the data
 - motion-compensated noise reduction, interpolation, or video segmentation

3.11 Motion Detection and Estimation - Motion Estimation

- Develop a motion estimation algorithm, three important elements need to be considered: models, estimation criteria and search strategies
- It cannot be considered a universal classification scheme of motion estimation algorithms, but it is very useful in understanding the properties and merits of various approaches.

3.11 Motion Detection and Estimation - Motion Estimation - Motion models

- Spatial Motion Models: estimate the 2D motion, or apparent motion, of image points
- Motion caused by the motion of 3D objects in a scene and the 3D motion of the camera
 - Camera motion affects the movements of all or almost all image points → compensated for by estimating or measuring it
 - 3D objects motion affects a subset of image points corresponding to objects' projections
- 3D objects motion depends
 - image formation model, e.g., perspective, orthographic projection
 - motion model of 3D object, e.g., rigid-body with 3D translation and rotation, 3D affine motion,
 - surface model of 3D object, e.g., planar, parabolic

3.11 Motion Detection and Estimation - Motion Estimation - Motion models

- Temporal motion models: The trajectories of individual image points drawn in the (x, y, t) space of an image sequence can be fairly arbitrary since they depend on object motion

$$\mathbf{x}(\tau) = \mathbf{x}(t) + \mathbf{v}_t(\mathbf{x}) \cdot (\tau - t) = \mathbf{x}(t) + \mathbf{d}_{t,\tau}(\mathbf{x})$$

- If there is an acceleration?
- Disadvantage
 - Require 2 parameters without acceleration and 4 with acceleration
- Advantage if combining spatial motion model and temporal motion model

3.11 Motion Detection and Estimation - Estimation Criteria – Pixel-Domain Criteria

- From the constant-intensity assumption it aims at the minimization of a function (e.g., absolute value) of the error

$$\varepsilon_k[\mathbf{n}] = I_k[\mathbf{n}] - \tilde{I}_k[\mathbf{n}]$$

- Motion fields calculated solely by minimization of the prediction error are sensitive to noise if the number of pixels in the region of support is not large compared to the number of motion parameters estimated or if the region is poorly textured.

- A more general estimation criteria $\varepsilon(\mathbf{d}) = \sum_{\mathbf{n} \in \mathcal{R}} \Phi(I_k[\mathbf{n}] - \tilde{I}_k[\mathbf{n}])$
- Φ is non-negative real-valued function

$$\begin{cases} \Phi(\varepsilon) = \varepsilon^2 & \text{not good: a single large error overcontributes to } \varepsilon \\ \Phi(\varepsilon) = \alpha|\varepsilon| & \text{more robust} \\ \Phi(\varepsilon) = \log(1 + \varepsilon^2/\omega^2) & \text{grows slower than } |x| \end{cases}$$

3.11 Motion Detection and Estimation - Estimation Criteria – f -Domain Criteria

- The shift property of the Fourier transform

$$\mathfrak{F} \{ I_{k-1}[\mathbf{n} - \mathbf{z}] \} = \hat{I}_{k-1}(\mathbf{u}) e^{-j2\pi \mathbf{u}^T \mathbf{z}}$$

- Argument difference

$$\arg \mathfrak{F} \{ I_k[\mathbf{n}] \} - \arg \mathfrak{F} \{ I_{k-1}[\mathbf{n}] \} = 2\pi \mathbf{u}^T \mathbf{z}$$

depends linearly on $\mathbf{z}$, the global motion can be recovered by evaluating the phase difference over a number of frequencies and solving the resulting over-constrained system of linear equations

3.12 Video Enhancement and Restoration

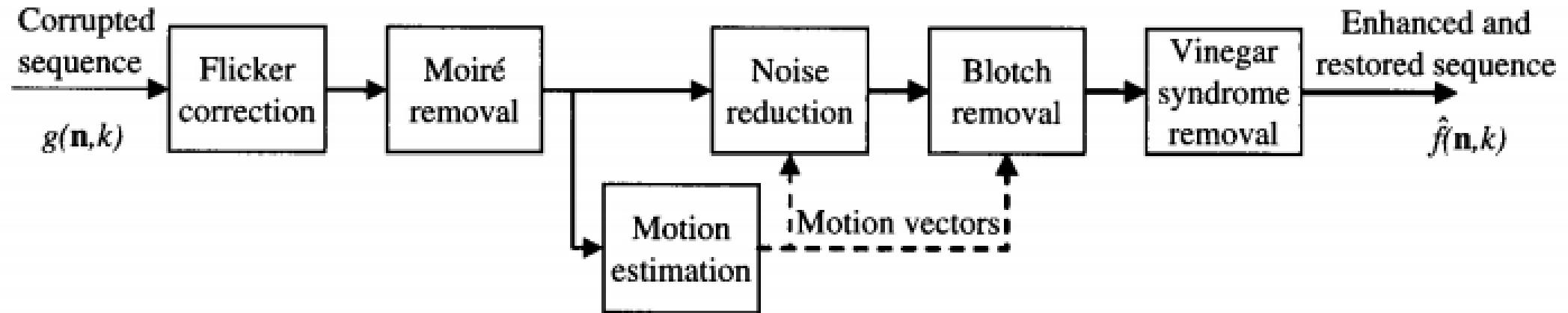
- Camera and digital recording technology, recorded video sequences can still suffer from severe quality degradations: imperfect or uncontrollable recording conditions, commonly encountered in fields like astronomy, forensic sciences, and medical imaging
- Video enhancement improves visual quality or enhance the performance of subsequent tasks such as analysis and interpretation
- Video enhancement and restoration preserve motion pictures and video tapes recorded over the last century
- The process involves transferring the original content to digital media, removing degradations to improve visual quality, and restoring the original spatial and temporal correlation.

3.12 Video Enhancement and Restoration

- key difference between enhancing and restoring 2D images versus video: the amount of data to be processed
- elaborate processing is feasible for individual images, it becomes impractical for the large volumes of data in medical sequences and film/video archives => enhancement and restoration methods for image sequences need to be of manageable complexity and semi-automatic

3.12 Video Enhancement and Restoration

- Some processing steps in the removal of various video artifacts.



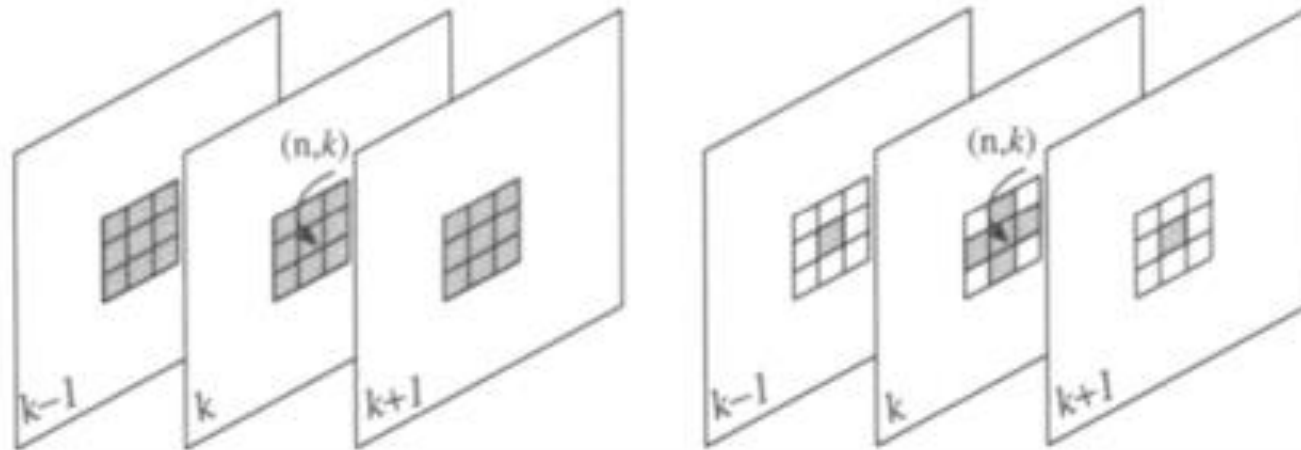
- Linear filter

$$\hat{f}(\mathbf{n}, k) = \sum_{l=-K}^K h(l)g(\mathbf{n}, k - l)$$

3.12 Video Enhancement and Restoration

- Order-Statistic Filters

$$\hat{f}(\mathbf{n}, k) = \sum_{r=1}^{|S|} h_r(\mathbf{n}, k) g_r(\mathbf{n}, k)$$



Summary

- A model of image restoration proceeds
- Noise models
- Spatial filtering and frequency domain filtering in noise reduction
- Degradation and its function
- Wavelet Denoising for Image Enhancement
- Minimum mean square error filtering
- Constrained least square filtering
- Basic Methods for Image Restoration and Identification
- Image reconstruction from projections
- Multichannel Image Recovery